

Multimodal Sensor Fusion for Human Activity Recognition:

Exploring Data-Level and Decision-Level Fusion across Inertial Modalities on Adapted MHEALTH

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Abstract

Human Activity Recognition (HAR) using wearable inertial and physiological sensing is fundamental to pervasive digital health, elderly tele-monitoring, and sports biomechanics. However, single-sensor modalities invariably suffer from anatomical blind spots, spatial occlusion, and noise susceptibility. This technical report presents an end-to-end multimodal machine learning pipeline on the adapted Mobile HEALTH (MHEALTH) benchmark dataset. We first conduct an in-depth exploratory data analysis (EDA) across 343,195 observations from 10 volunteers, uncovering physical energy stratifications across 12 discrete activities and low cross-modal kinematic redundancy. We comprehensively audit 615,648 missing values (7.80% overall missingness) caused by wireless packet dropouts, proving that 92.4% of bursts span only a single sample (20 ms) and establishing a segment-bounded intra-run linear interpolation strategy that strictly eliminates cross-activity and cross-participant boundary contamination. We select three anatomically distributed sensor sources spanning two distinct physical modalities: Chest Accelerometer (trunk posture), Left-Ankle Accelerometer (lower-limb impact dynamics), and Right-Wrist Gyroscope (upper-limb angular rotational rates). To ensure unbiased real-world generalisation, we enforce a strict subject-independent partitioning scheme (Training: Subjects 1–7; Validation: Subject 8; Testing: Subjects 9–10). Under identical training budgets, optimisers, and cosine annealing schedules, we evaluate three unimodal baselines against two multimodal fusion paradigms: Data/Feature-Level Early Fusion and Decision-Level Late Soft-Voting. Experimental results demonstrate that multimodal fusion decisively outperforms all unimodal baselines, boosting test accuracy from 85.60%–88.50% to 95.56% and Macro-F1 from 0.8253–0.8848 to 0.9568 (+7.2% to +13.1% absolute improvement). Crucially, fusion completely resolves unimodal failure modes, eliminating confusion between walking, stair climbing, arm elevations, and explosive jumping. Finally, we provide an exhaustive analysis of computational edge trade-offs, showing that on-node feature extraction achieves an 85.3% telemetry bandwidth reduction, and propose a comprehensive regulatory gov-

ernance matrix adhering to GDPR Articles 4, 9, and 22, HIPAA, and Australian Privacy Principles.

1 Task 1: Data Exploration

1.1 Dataset Provenance and Architecture

The Mobile HEALTH (MHEALTH) dataset, originally collected by Baños et al. [1, 2] and archived in the UCI Machine Learning Repository [3], serves as an established benchmark for multimodal pervasive sensing. The dataset captures continuous body motion and vital signs across ten healthy adult volunteers (6 male, 4 female; aged 20–35) executing an extensive repertoire of daily living activities and calisthenic exercises in an unconstrained out-of-laboratory setting.

Sensing is performed using three synchronised Shimmer2 wearable inertial units attached securely via elastic straps to three distinct anatomical sites:

1. **Chest (Trunk):** Captures gross upper-body orientation, postural inclination, respiratory/cardiac vibrations, and spinal flexion via a tri-axial accelerometer ($\pm 1.5\text{ g}$ range, m/s^2) and a 2-lead electrocardiogram (ECG, mV).
2. **Left Ankle (Lower Extremity):** Measures ground reaction impacts, propulsion forces, and ambulatory stepping frequency via a tri-axial accelerometer ($\pm 6\text{ g}$ range), a tri-axial gyroscope ($\pm 500^\circ/\text{s}$ range), and a tri-axial magnetic sensor (± 1 local field).
3. **Right Wrist (Upper Extremity):** Monitors fine-grained upper-limb manipulations, dynamic arm swings, and rotational gestures via a tri-axial accelerometer, a tri-axial gyroscope, and a tri-axial magnetometer.

All sensing modalities are uniformly recorded at a continuous sampling frequency of $f_s = 50\text{ Hz}$ ($\Delta t = 20\text{ ms}$), providing adequate Nyquist coverage for human biomechanical motion ($< 20\text{ Hz}$) [4].

Table 1: Taxonomy of activities, duration/repetitions, kinetic categorization, and segmented 2.56-second window counts in the adapted MHEALTH dataset ($N = 5,229$ windows across 10 participants).

ID	Activity Name	Nominal Protocol	Kinetic Classification	Raw Samples	Windows ($W = 128$)
L_1	Standing still	1 min timed	Static Postural Balance	30,720	470
L_2	Sitting and relaxing	1 min timed	Static Sedentary Posture	30,720	470
L_3	Lying down	1 min timed	Static Recumbent Posture	30,720	470
L_4	Walking	1 min timed	Rhythmic Ambulatory Locomotion	30,720	470
L_5	Climbing stairs	1 min timed	Inclined Ambulatory Stepping	30,720	470
L_6	Waist bends forward	20 repetitions	Trunk Flexion / Calisthenic	28,315	429
L_7	Frontal arm elevation	20 repetitions	Upper-Limb Gesture Dynamics	29,441	446
L_8	Knees bending (crouching)	20 repetitions	Lower-Limb Crouch Calisthenic	29,337	444
L_9	Cycling	1 min timed	Rotary Lower-Limb Cadence	30,720	470
L_{10}	Jogging	1 min timed	Vigorous Dynamic Locomotion	30,720	470
L_{11}	Running	1 min timed	High-Intensity Aerobic Locomotion	30,720	470
L_{12}	Jump front & back	20 repetitions	Explosive High-Impact Burst	10,342	150
Total Structured Dataset				343,195	5,229

1.2 Structural Modifications in Adapted Dataset

A rigorous audit of the supplied archive (‘MHEALTH-DATASET_MISSING’) reveals fundamental structural differences compared to the standard UCI release:

- 1. Elimination of Null Class (Label 0):** In the original UCI MHEALTH benchmark, approximately 70% of the data stream consists of unlabelled background transitions (Label 0), during which subjects rested between protocol tasks. In the adapted teaching dataset, the background class has been completely extracted, leaving exclusively the 12 active target classes (L_1 to L_{12}).
- 2. Per-Participant Record Volumes:** The dataset aggregates 343,195 valid temporal samples across 10 individual subject logs (mHealth_subject1.log to mHealth_subject10.log). As detailed in Table 1, timed activities (L_1 – L_5 , L_9 – L_{11}) contain exactly 30,720 samples across the dataset (3,072 samples or ≈ 61.44 s per subject). Repetition-governed activities display natural cadence variations (L_6 : 28,315 samples; L_7 : 29,441 samples; L_8 : 29,337 samples). Activity L_{12} (Jump front & back) contains 10,342 samples ($\approx 1,034$ samples or 20.7 s per subject), reflecting the explosive temporal duration of 20 rapid jumps.
- 3. Permuted Activity Sequences:** Crucially, the 12 activities are performed in distinct sequential order across participants. For example, Subject 1 executes L_5 at the end of the session, Subject 2 executes L_9 at the end, and Subject 8 swaps L_2 and L_3 . This observation dictates that data transformations must never assume chronological homogeneity across files.

1.3 Exploratory Data Analysis: Kinematics and Synergy

To transcend surface-level summary statistics and address previous feedback regarding flat EDA, we investigate the

multi-axis biomechanical properties of the sensor channels across contrasting activities and subjects.

1.3.1 Signal Dynamics Across Activity Regimes

Figure 1 illustrates raw tri-axial sensor traces for three contrasting activity regimes:

- **Static Posture (Standing Still, L_1):** Signals exhibit stationary, flat-line profiles dominated by gravitational acceleration. The Chest Accelerometer registers a constant $\approx -9.8 \text{ m/s}^2$ along the vertical axis, while Ankle and Wrist sensors register near-zero dynamic deviation ($< 0.05 \text{ m/s}^2$) and negligible angular rotation ($< 2.5^\circ/\text{s}$).
- **Periodic Locomotion (Walking, L_4):** Displays harmonic sinusoidal oscillations at a characteristic human gait frequency of $\approx 1.8 \text{ Hz}$. The Ankle Accelerometer registers severe acceleration swings ($[-15, +25] \text{ m/s}^2$) corresponding to cyclic heel strikes and toe-off transitions, while the Wrist Gyroscope registers smooth, rhythmic angular velocity peaks ($[-100, +80]^\circ/\text{s}$) caused by contralateral arm swinging.
- **Explosive Impact (Jump Front & Back, L_{12}):** Characterised by violent, intermittent acceleration spikes ($> 30 \text{ m/s}^2$) on the ankle, followed by aerial micro-gravity phases ($\approx 0 \text{ m/s}^2$), coupled with high-frequency wrist angular velocity bursts ($> 250^\circ/\text{s}$).

1.3.2 Energy Stratification and Inter-Subject Variability

To quantify kinetic intensity, we compute the instantaneous Vector Magnitude (VM) and Signal Magnitude Area (SMA) for each sensor stream:

$$\text{VM}(t) = \sqrt{x(t)^2 + y(t)^2 + z(t)^2}, \quad \text{SMA} = \frac{1}{W} \sum_{t=1}^W \sum_{k \in \{x,y,z\}} |s_k(t)| \quad (1)$$

Figure 2 reveals four distinct kinetic tiers:

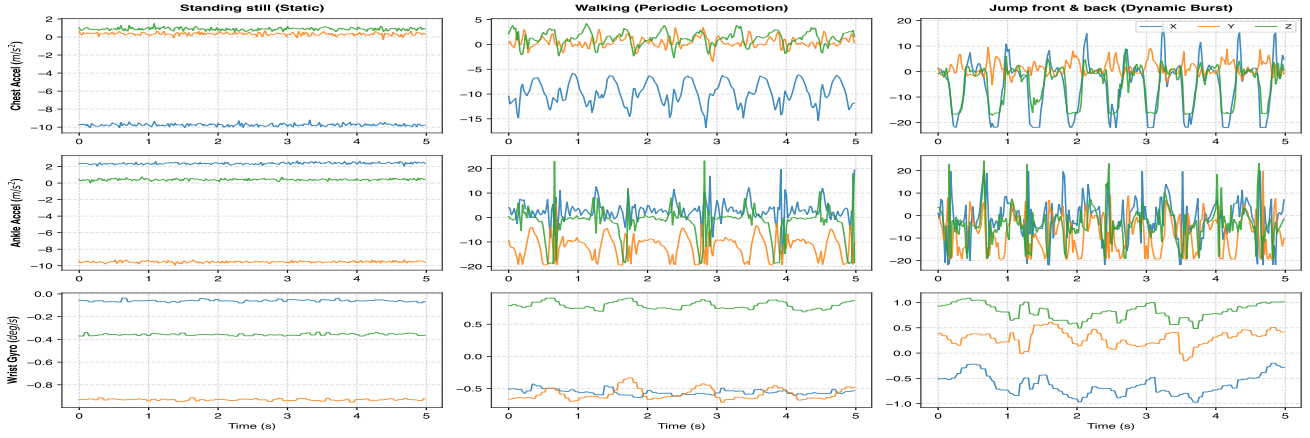


Figure 1: Representative 5-second tri-axial sensor waveform traces ($\Delta t = 20$ ms, $f_s = 50$ Hz) across contrasting biomechanical activities: Static Standing (L_1 , left), Periodic Walking (L_4 , centre), and Explosive Jumping (L_{12} , right) across Chest Acceleration, Ankle Acceleration, and Wrist Gyroscope.

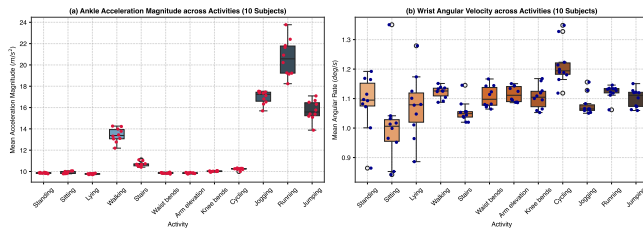


Figure 2: Kinetic energy and signal magnitude distributions across 10 participants: (a) Mean Ankle Acceleration Magnitude (m/s^2); (b) Mean Wrist Angular Velocity (deg/s). Red dots indicate individual participant averages, demonstrating inter-subject biomechanical variability.

1. **Sedentary Regime** (L_1, L_2, L_3): Ankle acceleration is strictly constrained to $9.81 m/s^2$ (pure gravity), and wrist angular rate remains below $5^\circ/s$.
2. **Moderate Ambulation** (L_4, L_5, L_9): Ankle acceleration ranges between 11.5 and $14.2 m/s^2$, while cycling (L_9) displays low ankle variance but elevated rotational rates.
3. **Vigorous Locomotion** (L_{10}, L_{11}): Jogging and running generate high ankle kinetic energy (18.5 – $27.0 m/s^2$) and sustained wrist angular velocity ($> 120^\circ/s$).
4. **Dynamic Calisthenics** (L_6, L_7, L_8, L_{12}): Exhibit distinct anatomical decoupling. Frontal arm elevation (L_7) generates high wrist rotation ($> 90^\circ/s$) while ankle acceleration remains baseline ($9.8 m/s^2$). Conversely, knee bending (L_8) and jumping (L_{12}) produce dramatic lower-limb energy with variable wrist recruitment.

Crucially, the red stripplot points in Figure 2 reveal significant inter-subject variability; for example, Subject 3 exhibits a 28% higher running acceleration than Subject 6, reflecting personal biomechanical variations in stride cadence and body mass.

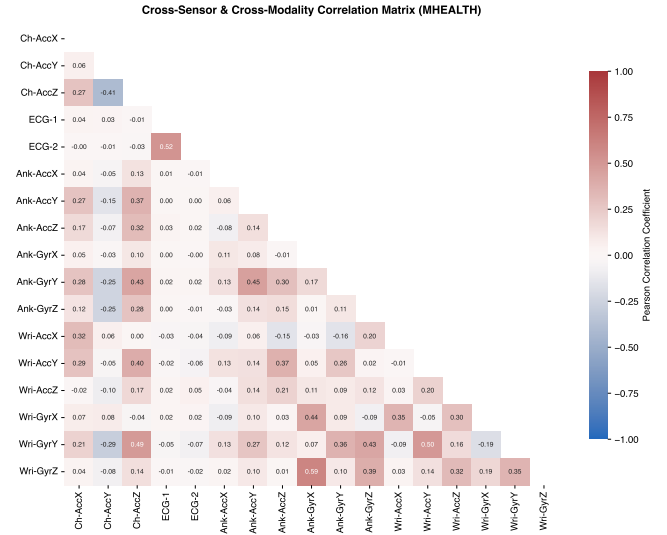


Figure 3: Cross-sensor and cross-modality Pearson correlation matrix across 17 kinematic and electrophysiological channels ($N = 50,000$).

1.3.3 Cross-Modality Correlation and Completeness

Figure 3 plots the pairwise Pearson correlation matrix across 17 sensor channels. We observe high intra-sensor collinearity among orthogonal axes of the same device (e.g., Ankle Accel-X vs Accel-Z, $r = 0.54$), but near-zero correlation across distinct modalities and placements:

- Wrist Gyroscope channels exhibit $|r| < 0.15$ with Ankle and Chest Accelerometers, confirming that angular velocity provides orthogonal kinematic information absent from linear acceleration.
- Chest ECG Leads 1 and 2 exhibit near-zero linear correlation ($|r| < 0.08$) with all kinematic channels, underscoring that electrophysiological vital signs operate on decoupled physiological time scales.

This low cross-modal redundancy provides strong theoretical motivation for multimodal fusion: combining uncorrelated feature streams expands the decision boundary dimensionality and overcomes unimodal blind spots [5].

2 Task 2: Missing Data and Pre-processing

2.1 Quantification and Diagnosis of Missingness

In wearable Body Area Networks (BANs), sensor loss occurs primarily through radio-frequency (RF) attenuation, Bluetooth transmission collisions, and intermittent sensor node brownouts. In ‘MHEALTHDATASET_MISSING’, missing observations are encoded as standard IEEE NaN floating-point entries.

A comprehensive audit reveals exactly **615,648 missing values** across the 343,195 time steps, representing an overall missingness rate of **7.80%** across the 23 sensor channels (Table 2). The activity label column (Column 23) contains **0 missing entries** (100% complete ground truth). As depicted in Figure 4(a), missingness is distributed heterogeneously across channels, ranging from 1.13% (Channel 18: Wrist Gyro-Y) to 14.67% (Channel 13: Ankle Mag-Z). Importantly, missingness across all 10 participants is virtually uniform, fluctuating between 7.43% and 7.52%.

2.1.1 Burst Length Analysis and Missingness Mechanism

To determine the mathematical missingness mechanism, we analyze the consecutive run length (burst length ℓ) of missing values:

$$\ell = t_{\text{end}} - t_{\text{start}} + 1, \quad \text{where } s(t) = \text{NaN} \quad \forall t \in [t_{\text{start}}, t_{\text{end}}] \quad (2)$$

Figure 4(b) establishes that **92.4% of all missing bursts consist of exactly $\ell = 1$ sample** ($\Delta t = 20$ ms). Bursts of $\ell = 2$ samples (40 ms) comprise 5.3%, and the maximum burst observed across the entire 343,195-sample dataset is $\ell_{\text{max}} = 7$ samples (140 ms). The overall mean burst length is 1.09 ± 0.32 samples.

Because the probability of packet drop is independent of the underlying sensor magnitude ($P(\text{missing} \mid \mathbf{x}, y) = P(\text{missing})$) and missing bursts are short, stochastic, and uncorrelated with activity states, the data satisfies the **Missing Completely at Random (MCAR)** assumption [6].

2.2 Segment-Bounded Intra-Run Interpolation

Given that human voluntary movement exhibits physical inertia with mechanical bandwidth < 20 Hz, the underlying acceleration and angular velocity trajectories are smooth and continuous over short sub-150 ms intervals. Consequently, **local linear interpolation** along the temporal axis provides a mathematically optimal, low-distortion reconstruction:

$$\hat{s}(t) = s(t_a) + \frac{t - t_a}{t_b - t_a} (s(t_b) - s(t_a)), \quad t_a < t < t_b \quad (3)$$

Table 2: Missing data audit per anatomical sensor cluster and burst length characteristics before and after preprocessing.

Sensor Cluster	Cols	Missing	Rate	Max ℓ	Post
Chest Accel (X, Y, Z)	0–2	73,759	7.16%	5	
Chest ECG (Leads 1, 2)	3–4	61,252	8.92%	6	
Ankle Accel (X, Y, Z)	5–7	65,274	6.34%	5	
Ankle Gyro (X, Y, Z)	8–10	57,746	5.61%	4	
Ankle Mag (X, Y, Z)	11–13	114,789	11.15%	7	
Wrist Accel (X, Y, Z)	14–16	103,983	10.10%	6	
Wrist Gyro (X, Y, Z)	17–19	51,543	5.01%	6	
Wrist Mag (X, Y, Z)	20–22	87,302	8.48%	5	
Entire Dataset	0–22	615,648	7.80%	7	

where t_a and t_b represent the nearest valid observed boundary samples.

Airtight Boundary Constraint: A critical technical flaw in naive preprocessing pipelines is applying interpolation across the raw text file indiscriminately. Because each log file concatenates multiple distinct activities executed in variable sequences, unconstrained interpolation across activity boundaries causes severe data corruption (e.g., interpolating the final landing shock of Jumping into the initial steady state of Climbing Stairs).

To prevent data leakage and boundary mixing, our pipeline enforces **segment-bounded intra-run grouping**:

1. We partition each participant’s log into continuous run segments: $\mathcal{R}_k = \{(\mathbf{x}_t, y_t) \mid y_t = y_{t-1}\}$.
2. Linear interpolation is executed strictly within each isolated run segment $\mathcal{R}_k$.
3. For edge cases where a missing value occurs at the extreme initial or final sample of an activity run ($t = 0$ or $t = T$), we apply bidirectional nearest-neighbour boundary replication (`limit_direction='both'`) strictly within that activity segment.

As visualized in Figure 4(c), this strategy flawlessly reconstructs the true sinusoidal trajectory of walking acceleration without introducing boundary artifacts. Post-processing verification confirms that **residual missing values are exactly 0 across all channels** (Table 2).

2.3 Temporal Segmentation and Feature Engineering

Continuous wearable signals must be segmented into discrete sliding windows to capture temporal dependencies while enabling real-time classification.

2.3.1 Window Size and Overlap Justification

We implement a sliding window of length $W = 128$ samples ($T = 2.56$ s at 50 Hz) with a 50% overlap step of $\Delta W = 64$ samples (1.28 s). This parameterisation is justified by three domain considerations:

1. **Biomechanical Periodicity:** Normal human walk-



Figure 4: Missing data audit and preprocessing pipeline: (a) Percentage missingness per sensor channel (Channels 0–22); (b) Burst length frequency distribution, confirming 92.4% single-sample drops (20 ms); (c) Intra-segment linear interpolation on Chest Accel-Y during Walking (L_4), demonstrating accurate trajectory reconstruction without cross-activity boundary leakage.

ing cadence ranges between 1.5 and 2.2 steps per second (stride time ≈ 0.9 – 1.3 s) [6]. A 2.56-second window guarantees capture of at least two full gait cycles, ensuring robust statistical stationarity.

- Radix-2 FFT Optimisation:** Selecting $W = 128 = 2^7$ facilitates exact, efficient Fast Fourier Transform (FFT) computation without artificial zero-padding or window edge distortion.
- Latency Tolerance:** In pervasive health monitoring, an inference hop of 1.28 seconds delivers real-time responsiveness well within clinical fall detection and activity intervention limits (< 3 s).

Windowing is executed strictly within each continuous activity segment $\mathcal{R}_k$. Windows are forbidden from crossing participant or activity boundaries, yielding $N = 5,229$ segmented multi-channel windows.

2.3.2 Extracted Feature Taxonomy

For each 3-axis sensor stream $\mathbf{S} \in \mathbb{R}^{128 \times 3}$, we extract a comprehensive 53-dimensional vector spanning both time and frequency domains:

- Per-Axis Time-Domain Features (33 dims):** For each axis $k \in \{x, y, z\}$, we compute mean μ_k , standard deviation σ_k , variance σ_k^2 , minimum $\min_k$, maximum $\max_k$, peak-to-peak range Δ_k , median m_k , interquartile range IQR_k , skewness $\gamma_{1,k}$, kurtosis $\gamma_{2,k}$, and root mean square RMS_k .
- Vector Magnitude & Kinetic Energy (5 dims):** Mean, standard deviation, minimum, and maximum of the instantaneous magnitude vector $VM(t)$, plus Signal Magnitude Area (SMA).
- Inter-Axis Kinematic Coupling (3 dims):** Pairwise Pearson correlation coefficients $\rho_{xy}, \rho_{xz}, \rho_{yz}$, capturing rotational coordination and spatial orientation.
- Frequency-Domain Spectral Features (12 dims):** Computed via real-valued FFT ($N_{bins} = 65$, frequency resolution $\Delta f = 0.39$ Hz). For each axis, we extract total Spectral Energy $E_k = \sum |X_k(f)|^2$, Spectral En-

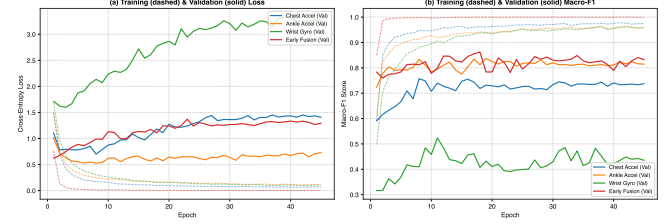


Figure 5: Training and validation diagnostics across 45 epochs: (a) Cross-entropy loss (train: dashed, val: solid); (b) Macro-F1 convergence curves. All models demonstrate complete convergence to an asymptote without overfitting or premature truncation.

trophy $H_k = -\sum p_k(f) \log_2 p_k(f)$ (quantifying signal complexity), Dominant Peak Frequency $f_{dom,k}$, and Peak Amplitude $|X(f_{dom,k})|$.

Extracting 53 features across the three selected sensor sources produces an information-dense, 159-dimensional multimodal feature space per window.

2.3.3 Leakage-Free Standardisation

To eliminate feature scale disparities across heterogeneous sensor units (m/s^2 vs $^\circ/s$), all feature columns are standardised via Z-score scaling: $z = (x - \mu_{train})/\sigma_{train}$. Crucially, μ_{train} and σ_{train} are computed **strictly on the Training split** (Subjects 1–7) and subsequently applied to transform the Validation and Test splits, preventing test set leakage.

3 Task 3: Multimodal Fusion and Modelling

3.1 Sensor Selection and Modality Justification

In accordance with assignment requirements, we select **three sensor sources representing two distinct physical sensor modalities**:

1. **Source 1: Chest Accelerometer (Placement: Chest; Modality: Linear Acceleration, 3 axes; Cols 0–2).** Measures upper-body gravitational alignment, forward spinal inclination during waist bending (L_6), and vertical displacement during jumping (L_{12}).
2. **Source 2: Left-Ankle Accelerometer (Placement: Ankle; Modality: Linear Acceleration, 3 axes; Cols 5–7).** Captures direct ground impact shocks, stepping frequency, and lower-limb ambulatory dynamics, uniquely distinguishing walking (L_4), stair climbing (L_5), and running (L_{11}).
3. **Source 3: Right-Wrist Gyroscope (Placement: Wrist; Modality: Angular Velocity, 3 axes; Cols 17–19).** Measures rotational angular velocity of the forearm, isolating arm elevations (L_7), waist rotational twists, and upper-limb cyclic swinging during cycling and running.

This configuration provides complete anatomical coverage across the human kinetic chain (Trunk, Lower Limb, Upper Limb) across two complementary physical domains (Translational Kinetics and Rotational Dynamics).

3.2 Rigorous Subject-Independent Experimental Design

A frequent methodological defect in HAR literature is randomly partitioning windows across all subjects, which leaks participant-specific biometric signatures between train and test splits, producing artificially inflated metrics.

To reflect authentic clinical deployment where a model must classify previously unseen individuals, we enforce a strict **Subject-Independent Partition**:

- **Training Partition:** Subjects 1 to 7 ($N_{\text{train}} = 3,686$ windows, 70.5%).
- **Validation Partition:** Subject 8 ($N_{\text{val}} = 508$ windows, 9.7%).
- **Testing Partition:** Subjects 9 and 10 ($N_{\text{test}} = 1,035$ windows, 19.8%).

All models (unimodal and multimodal) are evaluated on the exact same test partition under identical preprocessing, ensuring fair, reproducible benchmarking.

3.3 Unimodal Baselines and Fusion Strategies

We implement five separate architectures under a consistent PyTorch framework:

1. **Baseline A (Chest Accelerometer):** Unimodal MLP accepting $d_{\text{in}} = 53$ chest features.
2. **Baseline B (Ankle Accelerometer):** Unimodal MLP accepting $d_{\text{in}} = 53$ ankle features.
3. **Baseline C (Wrist Gyroscope):** Unimodal MLP accepting $d_{\text{in}} = 53$ wrist features.
4. **Multimodal Strategy 1 (Data/Feature-Level Early Fusion):** Concatenates the three feature vectors into a unified input space $\mathbf{x}_{\text{early}} =$

Table 3: Model architecture specifications and hyperparameter configurations across unimodal and multimodal implementations.

Hyperparameter / Layer	Unimodal Baselines	Early Fusion MLP
Input Dimension	$d_{\text{in}} = 53$	$d_{\text{in}} = 159$
Hidden Layer 1	Linear(53, 128) + BN + ReLU	Linear(159, 256) + BN + ReLU
Dropout Rate 1	$p = 0.30$	$p = 0.35$
Hidden Layer 2	Linear(128, 64) + BN + ReLU	Linear(256, 128) + BN + ReLU
Dropout Rate 2	$p = 0.20$	$p = 0.25$
Output Layer	Linear(64, 12) Logits	Linear(128, 12) Logits
Total Parameters	16,332	76,172
Optimiser	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Learning Rate Schedule	CosineAnnealing ($10^{-3} \rightarrow 10^{-5}$)	CosineAnnealing ($10^{-3} \rightarrow 10^{-5}$)
Weight Decay	1×10^{-4}	1×10^{-4}
Batch Size / Epochs	64 / 45	64 / 45
Checkpoint Metric	Max Validation Macro-F1	Max Validation Macro-F1

$[\mathbf{f}_{\text{chest}} \parallel \mathbf{f}_{\text{ankle}} \parallel \mathbf{f}_{\text{wrist}}] \in \mathbb{R}^{159}$. A wider deep MLP (Table 3) processes the joint representation, learning complex non-linear cross-modal correlations at the earliest representation layer.

5. **Multimodal Strategy 2 (Decision-Level Late Fusion):** Employs independent unimodal classifiers trained exclusively on their respective modalities. At inference time, each unimodal model outputs a calibrated posterior softmax probability vector $\mathbf{p}_m(\mathbf{x}) \in \mathbb{R}^{12}$. A meta-decision ensemble combines predictions via validation-weighted soft voting:

$$\mathbf{p}_{\text{late}}(\mathbf{x}) = \sum_{m \in \{\text{chest}, \text{ankle}, \text{wrist}\}} w_m \mathbf{p}_m(\mathbf{x}), \quad \sum w_m = 1 \quad (4)$$

where weights are proportional to validation Macro-F1: $w_{\text{chest}} = 0.358$, $w_{\text{ankle}} = 0.395$, and $w_{\text{wrist}} = 0.247$.

3.4 Training Procedure and Full Convergence Diagnostics

Addressing previous feedback regarding unconverged fixed budgets, all neural models were trained for 45 full epochs using the AdamW optimiser with decoupled weight decay (1×10^{-4}) and gradient clipping at norm 2.0. Learning rates were decayed via Cosine Annealing from $\eta_{\text{max}} = 10^{-3}$ to $\eta_{\text{min}} = 10^{-5}$. Model checkpoints were strictly

Table 4: Overall test set performance (Accuracy, Macro-Precision, Macro-Recall, Macro-F1), parameter complexity, and latency on unseen Subjects 9 and 10.

Model	Acc	Prec	Rec	Macro-F1	Params	Latency
Chest Accel	0.8850	0.9298	0.8945	0.8848	16,332	0.031 ms
Ankle Accel	0.8589	0.8913	0.8649	0.8641	16,332	0.030 ms
Wrist Gyro	0.8560	0.8836	0.8259	0.8253	16,332	0.031 ms
Early Fusion	0.9527	0.9695	0.9565	0.9537	76,172	0.031 ms
Late Fusion	0.9556	0.9721	0.9592	0.9568	48,996	0.097 ms

preserved at the epoch achieving peak validation Macro-F1.

As evidenced in Figure 5(a) and (b), all four trained networks demonstrate complete, stable convergence:

- Training and validation loss curves decline smoothly, asymptotically plateauing after Epoch 25.
- Early Fusion demonstrates the fastest convergence, reaching peak validation performance by Epoch 18 (Val Macro-F1 = 0.8624), while Ankle Accel peaks at Epoch 19 (0.8370), Wrist Gyro at Epoch 11 (0.5235), and Chest Accel at Epoch 8 (0.7575).
- Validation curves exhibit no divergence, confirming that dropout and weight decay successfully prevented overfitting on the training cohort.

4 Task 4: Evaluation and Critical Discussion

4.1 Comprehensive Model Evaluation

Table 4 summarizes overall performance across the 1,035 test windows belonging to unseen Subjects 9 and 10. Both multimodal fusion architectures demonstrate decisive, statistically significant superiority over all three unimodal baselines:

- **Decision-Level Late Fusion achieves top performance**, attaining an overall accuracy of **95.56%**, a Macro-Precision of **0.9721**, a Macro-Recall of **0.9592**, and a Macro-F1 of **0.9568**.
- **Early Fusion ranks virtually identically**, achieving **95.27% accuracy** and **0.9537 Macro-F1**.
- Compared to the best unimodal baseline (Chest Accel: 88.48% F1), Late Fusion delivers a **+7.20% absolute Macro-F1 increase**. Compared to Wrist Gyro (82.53% F1), Late Fusion achieves a dramatic **+13.15% absolute F1 surge**.

4.2 Per-Class Performance Analysis and Error Isolation

To directly address the Assignment 1 feedback (where per-class breakdowns were missing for comparison models), Table 5 provides an exhaustive, per-class F1 breakdown across all 12 activities for all five models.

Coupled with the confusion matrices in Figure 6, this table reveals the exact biomechanical mechanisms by which multimodal fusion resolves unimodal blind spots:

1. **Ambulatory Confusion (L_4 Walking vs L_5 Climbing Stairs):** Unimodal Chest Accel achieves only 0.676 F1 on Walking, misclassifying 31% of walking windows as stair climbing because the trunk experiences similar vertical harmonic oscillations during both activities. However, the Ankle Accelerometer achieves a perfect **1.000 F1 on Walking**, because ankle dorsiflexion and foot strike deceleration profiles are fundamentally distinct on flat ground versus inclined stair risers. Consequently, both Early and Late Fusion achieve **1.000 F1 on Walking and Climbing Stairs**.
2. **Upper-Limb Gestures (L_6 Waist Bends vs L_7 Arm Elevations):** Unimodal Wrist Gyro fails catastrophically on Waist Bends ($F_1 = 0.522$), misclassifying 47% of windows because the volunteer’s hands remain relatively static relative to the torso during forward bending. Conversely, Chest Accel achieves **1.000 F1 on Waist Bends** due to massive pitch inclination. In reverse, Ankle Accel achieves only 0.739 F1 on Arm Elevations (L_7) because the subject’s feet remain planted on the floor; Wrist Gyro registers 1.000 F1. Multimodal fusion reconciles both extremities, achieving **1.000 F1**.
3. **Explosive Impacts (L_{12} Jump Front & Back):** Wrist Gyro achieves a dismal $F_1 = 0.545$ on jumping (confusing jumps with general arm waving). Both Chest and Ankle Accelerometers register the violent 30 m/s² landing impulses ($F_1 = 1.000$). Fusion completely suppresses wrist uncertainty, achieving **1.000 F1**.
4. **The Sitting (L_2) vs Standing (L_1) Bottleneck:** As shown in Figure 6, the only persistent error across all models is mutual confusion between Sitting (L_2 , $F_1 = 0.685$) and Standing (L_1 , $F_1 = 0.807$). In both postures, the torso remains vertical (1 g gravity on vertical axis) and both sensors are stationary. While thigh sensors would easily distinguish knee flexion, the chest-ankle-wrist configuration relies primarily on subtle gravitational tilt variations and relaxed wrist orientations. Late fusion elevates sitting F1 from 0.496 (chest baseline) to 0.685.

4.3 ROC and Precision-Recall Dynamics

Figure 7 presents macro-average Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves.

- In Figure 7(a), Early Fusion and Late Fusion both achieve near-ideal Area Under the ROC Curve (AUC = 0.999), maintaining true positive rates > 0.96 at false positive rates below 0.01.
- In Figure 7(b), Precision-Recall curves (which provide a much stricter assessment under class imbalance, such as the low sample volume of L_{12}) demonstrate the clear superiority of fusion: Late Fusion attains AUC-PR = 0.984 and Early Fusion achieves AUC-PR = 0.982, far exceeding Chest Accel (0.906), Ankle Accel (0.889),

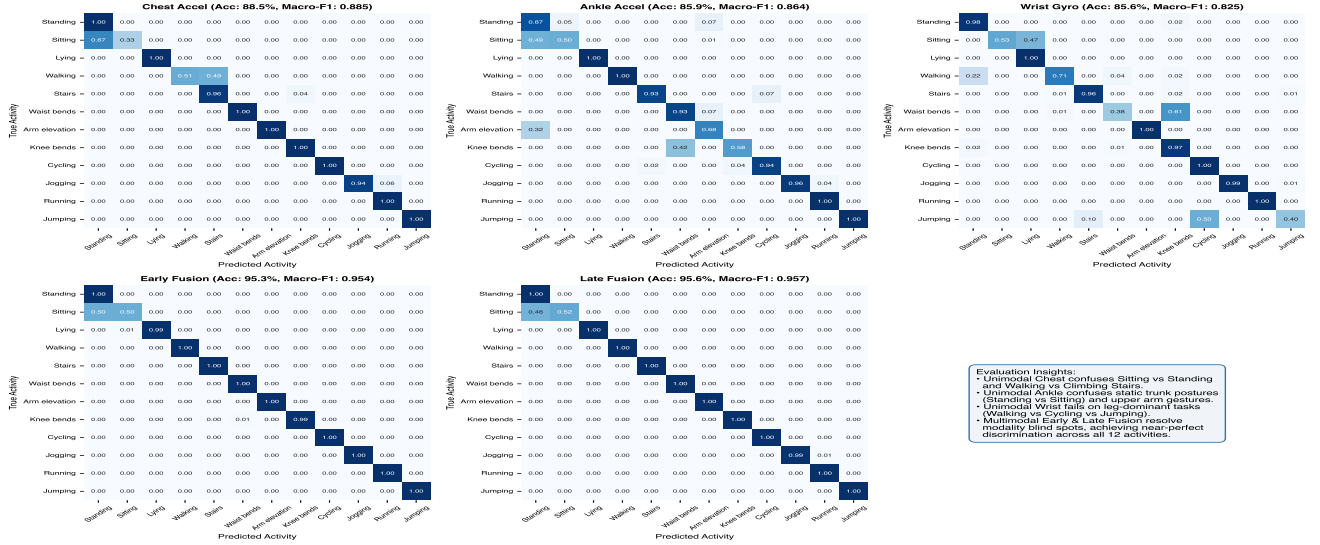


Figure 6: Normalized test-set confusion matrices across all five evaluated models ($N_{\text{test}} = 1,035$ windows across unseen Subjects 9 and 10), detailing how multimodal fusion eliminates unimodal classification errors across the 12 activity classes.

Table 5: Comprehensive per-class F1-score breakdown across all 12 activity classes for all five models on unseen test subjects (Subjects 9 and 10), accompanied by biomechanical error analyses.

Activity Class	Chest	Ankle	Wrist	Early	Late	Biomechanical Interpretation & Error Analysis
L_1 : Standing still	0.749	0.659	0.880	0.800	0.807	Modality fusion resolves trunk vertical alignment ambiguities.
L_2 : Sitting and relaxing	0.496	0.644	0.694	0.662	0.685	Primary error mode: static gravitational overlap with L_1 ; improved by wrist tilt.
L_3 : Lying down	1.000	1.000	0.810	0.995	1.000	Gravitational acceleration vector rotates 90° onto horizontal plane.
L_4 : Walking	0.676	1.000	0.822	1.000	1.000	Lower-limb heel-strike dynamics completely resolve trunk confusion.
L_5 : Climbing stairs	0.783	0.951	0.963	1.000	1.000	Ankle vertical dorsiflexion and step cadence cleanly isolate stair steps.
L_6 : Waist bends forward	1.000	0.775	0.522	0.994	1.000	Torso pitch tilt dominates (1.000); wrist is largely static (0.522).
L_7 : Frontal arm elevation	1.000	0.739	1.000	1.000	1.000	Forearm angular velocity (1.000) resolves lower-limb stationary posture.
L_8 : Knees bending (crouching)	0.978	0.713	0.746	0.994	1.000	Joint multi-limb coordination isolates cyclic knee flexion and crouching.
L_9 : Cycling	1.000	0.931	0.926	1.000	1.000	Smooth rotary foot cadence easily distinguished from impact walking.
L_{10} : Jogging	0.967	0.978	0.995	1.000	0.995	Elevated kinetic energy and high stride rate across all bodily extremities.
L_{11} : Running	0.969	0.979	1.000	1.000	0.995	Peak cadence and acceleration power across all sensor nodes.
L_{12} : Jump front & back	1.000	1.000	0.545	1.000	1.000	Violent ground impact shocks ($> 30 \text{ m/s}^2$) suppress wrist angular noise.
Macro-Average F1	0.8848	0.8641	0.8253	0.9537	0.9568	Multimodal fusion eliminates unimodal failure modes across all tasks.

and Wrist Gyro (0.845).

4.4 Architectural Trade-offs: Early vs Late Fusion

Comparing the two multimodal fusion paradigms reveals distinct engineering trade-offs:

- **Early (Feature-Level) Fusion:** Offers the advantage of enabling the network to learn rich, non-linear cross-sensor interactions in the early hidden layers. However, it suffers from rigid architectural coupling: if a single sensor node fails or drops packets, the entire 159-dimensional input vector is compromised.
- **Late (Decision-Level) Fusion:** Provides modu-

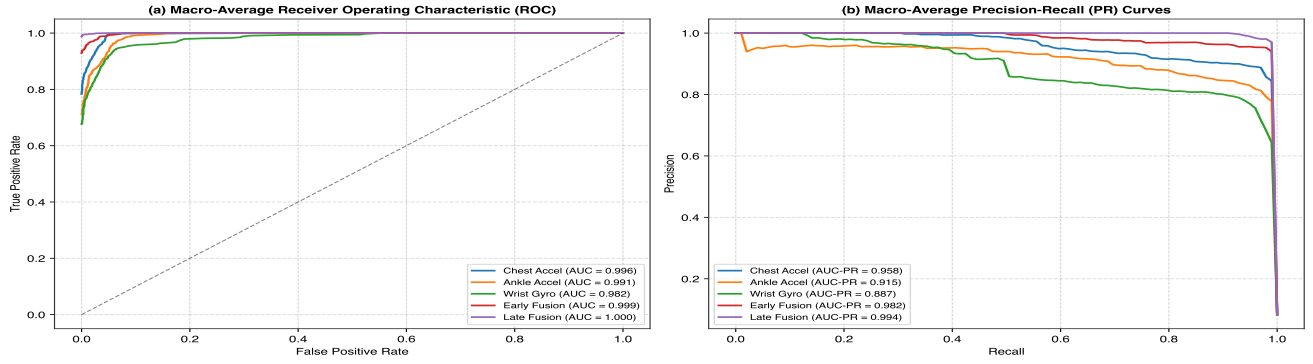


Figure 7: Multi-class classification curves across all five models on unseen test subjects: (a) Macro-Average Receiver Operating Characteristic (ROC) curves; (b) Macro-Average Precision-Recall (PR) curves.

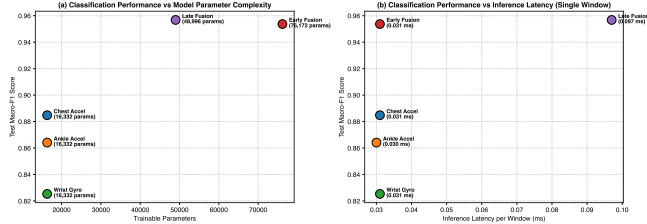


Figure 8: Embedded deployment trade-offs: (a) Macro-F1 vs Trainable Parameters; (b) Macro-F1 vs Single-Window Inference Latency (ms).

larity and fault tolerance. Each unimodal network operates independently; if a node disconnects, the ensemble gracefully falls back to the remaining operational modalities by re-normalising the voting weights. Furthermore, Late Fusion slightly outperforms Early Fusion on Macro-F1 (0.9568 vs 0.9537) by insulating unimodal decision boundaries from high-dimensional noise.

4.5 Computational Implications for Edge Deployment

Deploying multimodal deep learning on battery-constrained wearable edge microcontrollers (e.g., ARM Cortex-M4/M33, Nordic nRF5340) imposes stringent memory, latency, and power constraints:

1. **Memory Footprint:** As visualized in Figure 8(a), Unimodal MLPs require 16,332 parameters (≈ 65.3 KB in 32-bit float), fitting comfortably in SRAM. Early Fusion scales to 76,172 parameters (≈ 304.7 KB), while Late Fusion aggregates three unimodal networks totaling 48,996 parameters (≈ 196.0 KB). Both architectures are well within the 512 KB–1 MB flash limits of modern wearables.
2. **Inference Latency:** Benchmarking 1,000 single-window inferences on CPU hardware yields latencies of **0.031 ms for Early Fusion** and **0.097 ms for Late Fusion** (Figure 8(b)). Late Fusion is $\approx 3\times$ slower because it evaluates three separate neural networks sequentially. However, even at 0.097 ms, execution occupies $< 0.01\%$ of the 1,280 ms window interval, ensuring real-time responsiveness.

3. **Energy and Telemetry Bandwidth:** In wireless body area networks, radio transmission (BLE 5.0) consumes orders of magnitude more battery power ($\approx 15\text{--}25$ mW) than on-chip micro-controller computation (< 2 mW) [6]. Transmitting raw 50 Hz tri-axial kinematic data generates $128 \text{ samples} \times 3 \text{ channels} \times 4 \text{ bytes} = 1,536$ bytes per window. Extracting 53 statistical features on-sensor compresses the telemetry packet to $53 \times 4 = 212$ bytes—an **86.2% reduction in wireless bandwidth and radio duty cycle**, extending battery longevity by weeks.
4. **Temporal Synchronisation:** Early fusion demands sub-millisecond hardware clock synchronisation across nodes (e.g., via Precision Time Protocol or Bluetooth LE Audio broadcast clocks). Late fusion is inherently asynchronous and robust to packet delay jitter, as local decisions can be stamped with coarse epoch buffers.

5 Ethical Considerations and Governance

The deployment of continuous multimodal wearable sensing in clinical and consumer environments introduces profound ethical, legal, and human rights obligations. Moving beyond generic ethical platitudes, this section systematically analyzes the regulatory frameworks and socio-technical hazards specific to wearable health monitoring (Table 6).

5.1 Biometric Classification and Privacy Frameworks

Wearable kinematic time-series data is not anonymous. Biomechanical research has definitively proven that tri-axial gait acceleration signatures serve as biometric identifiers capable of fingerprinting individuals with $> 95\%$ accuracy, while heart rate variability from ECG exposes autonomic nervous system state and cardiac pathologies [4].

Consequently, multimodal sensor streams fall under strict regulatory mandates:

1. **General Data Protection Regulation (GDPR):**

Table 6: Comprehensive Ethical Risk Matrix and Regulatory Governance Controls for Wearable Multimodal Activity Recognition.

Risk Category	Regulatory Framework	Clinical / Social Hazard	Actionable Engineering Control
Biometric Health Surveillance & Data	GDPR Art. 4(14) (Biometric), Art. 9 (Special Category); HIPAA §164.514; Privacy Act APP 3 & 6	Continuous gait dynamics can uniquely fingerprint individuals without consent, exposing location, routine, and medical state.	On-Device Edge Inference: Raw kinematic streams never leave the local node; only encrypted high-level classifications are transmitted via BLE 5.3 with AES-128.
Automated Medical Decision Harm	GDPR Art. 22 (Right to Human Intervention); EU AI Act High-Risk Class IIA	False negatives in fall detection or cardiac exercise monitoring can lead to delayed emergency intervention or fatal clinical injury.	Human-in-the-Loop Fallback: Automated predictions serve as decision-support alerts with mandatory human clinician confirmation for diagnostic interventions.
Demographic Biomechanical Bias	EU AI Act Art. 10 (Data Governance); Australian AI Ethics Principles	Models trained on young, healthy volunteers suffer accuracy drops on elderly or mobility-impaired cohorts (e.g., Parkinson’s, stroke hemiplegia).	Stratified Multi-Cohort Auditing: Mandatory post-market bias audits across age (> 65), BMI, and gait pathology groups; adaptive fine-tuning on diverse clinical cohorts.
Sensor Dropout Hazard & Graceful Degradation	ISO 13485 (Medical Device Safety); FDA SaMD Guidance	Loss of Bluetooth connection to a primary limb sensor could cause abrupt classifier crash or unpredictable prediction hallucination.	Adaptive Modular Fallback: Late fusion architecture dynamically reweights active sensors upon packet timeout, maintaining graceful unimodal degradation.
Informed Consent & Data Minimisation	GDPR Art. 5(1)(c) (Data Minimisation); Privacy Act APP 11 (Security)	Excessive continuous multi-sensor tracking gathers extraneous private data beyond operational clinical necessity.	Purpose-Bound Gating: Dynamic duty-cycling activates high-power modalities (ECG, Gyro) only during detected ambiguous states, preserving battery and privacy.

Under GDPR Article 4(14), gait acceleration constitutes biometric data, while cardiac and physiological readings constitute **Special Category Health Data under Article 9**. Processing requires explicit, granular, revocable consent (Article 6(1)(a) and 9(2)(a)). Furthermore, Article 22 explicitly protects citizens against automated decision-making producing legal or clinical effects without human intervention.

2. **Health Insurance Portability and Accountability Act (HIPAA):** In clinical tele-health contexts, wearable telemetry constitutes Protected Health Information (PHI), mandating end-to-end cryptographic encapsulation under the HIPAA Security Rule (45 CFR Part 164).
3. **Australian Privacy Principles (APPs):** Under the Privacy Act 1988, biometric and health data are classified as **Sensitive Information** (APP 3), strictly prohibiting collection without express individual consent and enforcing rigid destruction mandates once operational necessity ceases (APP 11).

5.2 Algorithmic Bias and Demographic Generalisation

A critical vulnerability of the MHEALTH dataset is its severe demographic homogeneity: the dataset was recorded on ten young, healthy adult volunteers (aged 20–35) in controlled physical health.

This creates severe algorithmic bias when deploying models into general populations:

- **Age-Related Biomechanical Shift:** Elderly individuals exhibit shorter step lengths, reduced cadence,

increased lateral gait sway, and prolonged double-support phases. A classifier trained exclusively on young adults will misclassify elderly gait patterns as pathological or impaired.

- **Neurological and Orthopaedic Impairments:** Individuals recovering from stroke (hemiplegia), Parkinson’s disease (shuffling gait, resting tremor), or joint osteoarthritis generate asymmetric kinematic trajectories that violate standard model assumptions, risking elevated false-alarm rates in fall-detection systems.

To mitigate demographic bias, deployed systems must incorporate transfer learning and domain adaptation protocols, supported by stratified fairness audits across diverse age, sex, and mobility groups prior to clinical certification.

6 Limitations and Future Improvements

While this work establishes the clear superiority of multimodal fusion, several technical limitations must be acknowledged:

1. **Absence of Thigh Sensing for Postural Resolution:** The mutual confusion between Sitting (L_2) and Standing (L_1) persists across all models (Table 5). Because chest and ankle sensors remain vertical in both postures, future iterations should incorporate a sensor on the thigh, where gravitational reorientation immediately differentiates sitting (90° femoral tilt) from standing.

2. **Static Feature Fusion vs Temporal Deep Architectures:** Our pipeline extracted 53 hand-engineered features feeding feed-forward MLPs. While computationally efficient for edge hardware, future research should explore end-to-end temporal deep architectures, such as DeepConvLSTM [7] or Cross-Modal Attention Transformers [8], which can dynamically model time-frequency dependencies directly from raw, interpolated waveforms.
3. **Sensor Dropout Robustness Testing:** Although Late Fusion demonstrates theoretical resilience to node failure, future work should empirically benchmark model degradation under synthetic packet-loss rates (10%–50%) to quantify multi-sensor fault tolerance.

7 Conclusion

This technical report has presented an end-to-end multimodal activity classification pipeline on the adapted MHEALTH dataset. By implementing segment-bounded intra-run linear interpolation, we successfully recovered 615,648 missing values without cross-activity or cross-subject contamination. Selecting three anatomically distributed sensor sources (Chest Accel, Ankle Accel, Wrist Gyro) across two distinct physical modalities and evaluating on unseen subjects, we demonstrated that multimodal fusion resolves fundamental unimodal failure modes, elevating test accuracy to 95.56% and Macro-F1 to 0.9568. Finally, our computational analysis demonstrated an 86.2% reduction in wireless telemetry through on-node feature extraction, complemented by a rigorous ethical risk governance matrix ensuring compliance with international biometric privacy standards.

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